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Designing a low-cost eyeball tracking keyboard for paralyzed people☆

Zeenat Al-Kassim, Qurban Ali Memon*

EE Department, College of Engineering, UAE University, Al-Ain, 15551 United Arab Emirates

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ABSTRACT

This paper aims at designing a scanning keyboard that serves people with paralysis. This keyboard works on detection and tracking of the user eyeball movements. The main objective of this research work is to implement a simple and less expensive technique that produces the best possible results with a higher level of accuracy. The design of the system is done in phases; these are eye detection and tracking, followed by classification. The tracking phase incorporates skin color segmentation, black pupil detection and support vector machines. We also explored the role of eye red-channel, eye width and height for eye tracking. A scanning keyboard is developed and tested to work with three eyeball movements with 90% double detection accuracy. The performance of the algorithm is experimentally analyzed and the benefits of the proposed approach are highlighted. Also, the simulation results are presented.

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1. Introduction

Physical challenges emanate from problems and diseases such as spinal cord injuries, accidents, paralysis of various forms, etc. These problems cause a number of challenges to the sufferer, especially if he/she cannot communicate or move his/her body muscles. The number of such affected people is quite large in the Gulf Cooperation Council (GCC) countries, though the exact number is not known since research results in this direction are not up to the expectation of the public.

Over a number of decades, different solutions in the domain of support for physically challenged people have been suggested and put in practice, but they are not up to the mark as they lack in technology-assisted communication to enable automated mobility, etc. One such solution is the use of alternative ways to interact with devices for people with physical challenges. A Partner-Assisted Scanning (PAS), a form of Augmentative and Alternative Communication (AAC) is a well-known technique, where a nurse uses a keyboard in front of the physically challenged person to scan his/her eyes in selecting different characters or words to enable communication with the physically challenged.

The object tracking (both moving and stationary) is not a new field, and applications exist in various disciplines like engineering and healthcare. Conventionally, technology has been exploited to build systems that try to simplify human life. These systems utilize concepts like human-computer interaction, image processing, virtual/augmented reality, and so on, to build devices by which humans can communicate easily with the machines [1,2]. An example is the Kinect Sensor developed by Microsoft, and the Wii Remote developed by Nintendo Inc. However, these machines or tools help automate human-machine interaction to be used by normal individuals, while the physically challenged as addressed in this research

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* Corresponding author.
E-mail address: qurban.memon@uaeu.ac.ae (Q.A. Memon).

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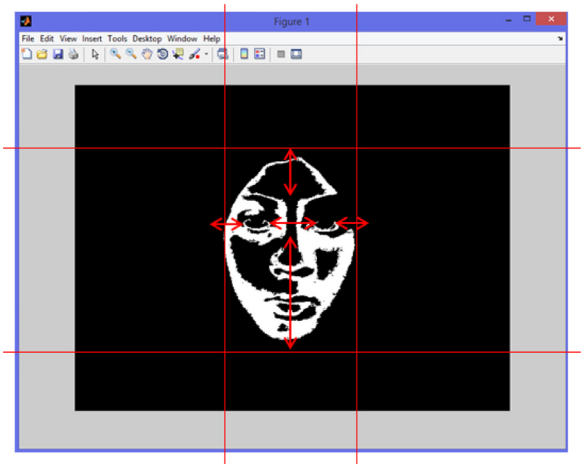


Fig. 1. Feature extraction (skin color, face measurements).

may be able to blink eye only once. Though algorithms exist in the literature for accurate measurements for eye detection and tracking, but such results are achieved once at least 66 frames are available for processing [3]. The challenge in eye tracking of the physically challenged is the availability of only a few frames as the physically challenged person may blink only once, which may provide approximately 20–25 frames, unlike other situations [3] where a significant number of frames is available to reduce errors over a series of frames.

The work in [4,5], though not exactly related to this work, exemplify real time image processing applications including healthcare. With recent technological developments, gesture recognition has the potential to introduce new ways of human-computer interaction and these could serve as a substitute for people with physical challenges. In this domain, the focus is on designing a system that is able to detect eyes of the user automatically. In that, one approach could be to train the system to detect the face of the user prior to detecting the user's eyes. When the face is detected, the eyes' location can be extracted from the face. This type of system can be developed for users with communication or mobility difficulties with the user assumed to have his/her head placed straight in front of the system screen. Head movements and tilting are not considered since the users targeted are with physical paralysis.

To achieve gesture recognition, researchers have used different approaches. In [6], the authors perform gaze estimation using the Scale Invariant Feature Transform (SIFT) along with Random Sample Concensus (RANSAC) and homography model. In another work [7], the authors apply the famous Viola-Jones classifiers for face detection, along with a mean shift tracking algorithm. With a different approach, the authors in [8] performed skin color segmentation for face detection, followed by circular Hough transform to detect the iris.

In this work, a low-cost algorithm is introduced, which investigates the role of red channel in the eye, black pupil, and eye height and width to track the center of the eye. This algorithm is experimentally analyzed and results are compared. In these experiments, data is acquired by a typical low-cost webcam (Logitech C110), where images are captured at 640-by-480 pixel resolution. The first phase of camera calibration or face detection is performed by a trained support vector machine (SVM), the details of which are discussed in a comparative experimental analysis in [9].

The paper is structured as follows. In the next section, eye tracking phase is discussed that outlines different steps needed to complete eyeball tracking. Section three presents classification phase with three user eyeball positions. Simulation results are presented in section four. In section five, benefits of the proposed system are discussed, followed by discussions and the concluding part in section six.

2. Eye tracking phase

In order to track eyes, calibration or eye detection phase is completed first, followed by feature (like skin color, facial measurements) extraction, as shown in Fig. 1.

In this phase, the face is detected by skin color segmentation, after which face coordinates are extracted. After detecting the face, eyes are tracked by two methods. In the first method, black pupil detection is performed, while in the second, eyes are detected by a trained SVM. If both methods detect eyes in the same region, a check is carried to make sure that both left and right eyes are horizontally inclined at the same angle. If yes, eyes are correctly tracked. As a summary, the steps of this algorithm are displayed in Fig. 2 with resulting corresponding visual displays in Fig. 3.

With reference to step 2 in Fig. 2, the face is detected by skin color segmentation, followed by extraction of face coordinates from the black and white image. These face coordinates refer to the width and height of the face with respect to the image size (typically found in every human face), as well as other measurements like the user's forehead area, chin area, and space occupied by the user's eyes. Using these facial measurements, the eye region is predicted as stated in Eqs. (1) and

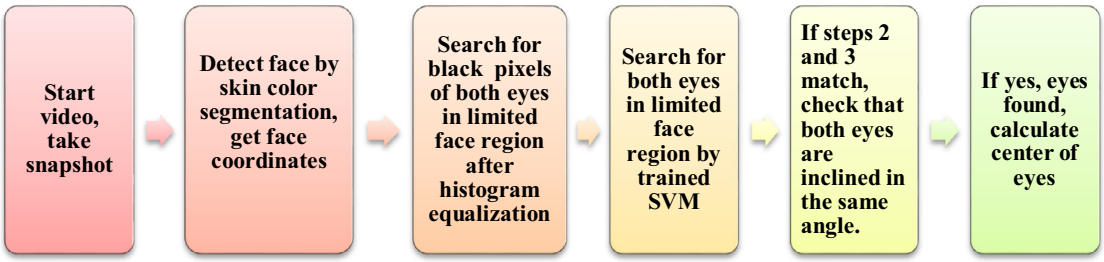


Fig. 2. Steps for tracking algorithm.

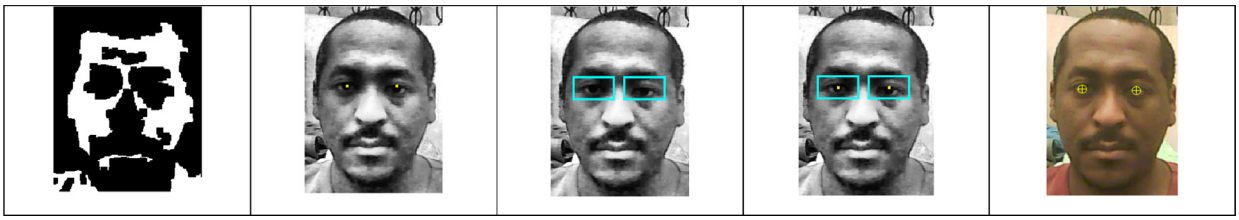


Fig. 3. Steps for tracking algorithm in pictures.



Fig. 4. Black pupil detection.

(2). In this predicted region, histogram equalization is performed, after which a search for the only black pupil is done using the Eq. (3), as shown in Fig. 4. This is because, it was found through trial and error experiment that black pupil search is more efficient than white cornea search, since bright pixels in other areas of the face may be mistaken as the white cornea due to reasons like surrounding lighting, etc. Using Eq. (3), we assume that the black pupil falls into the region of minimum pixel values (with range on a scale of 0–255).

$$Eye\ width = \frac{1}{4th}\ of\ face\ width \tag{1}$$

$$Eye\ height = 0.6\ \times Eye\ width \tag{2}$$

$$pixel\ value\ in\ range\ of\ (minimum \pm st.\ dev.\ of\ equalised\ red\ channel\ of\ the\ eye \tag{3}$$

In Eq. (3), contrast enhancement is applied to the red channel only. This is because it has been noticed that applying contrast enhancement on the red channel produces best results with regard to distinguishing between black and white pixels [10,11]. The reason behind this is the existence of red blood vessels behind the retina. Additionally, red color exhibits the highest wavelength compared to other colors, that is why, in [10,11], researchers preferred to work on the red color space represented in hue. For comparative purposes, Table 1 shows contrast enhancements on each channel after experimental testing.

With reference to step 3 in Fig. 2, the first step in training the SVM was to process the training eye samples. The training samples consist of both eye (represented by 1's) and non-eye (represented as −1's) samples. A kernel polynomial of degree three was selected because the training of linear and Gaussian didn't produce the desired minimum error rate and maximum correction rate, unlike polynomial kernel. An order three polynomial produced the minimum error rate with support vectors forming a margin that distinguishes between the eye and non-eye samples. Based on trial and error, different SVMs were trained by changing SVM parameters (i.e., value C, α) to obtain best results. The final SVM selected was trained with 100 samples (50 faces, 50 non-faces image), each sample resized to 35-by-65 (height-by-width). The 38 samples were selected as support vectors. After trial and error, constraint C was given a value of 0.00000001. After training, values of α were

Table 1
Histogram equalization on different color channels.

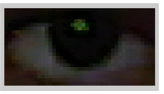
	Before histogram equalization	After histogram equalization
RGB channel		
Red channel		
Green channel		
Blue channel		



Fig. 5. Golden ratio in human face.

calculated by the kernel classifier, and the weight matrix was calculated by the Eq. (4).

$$w = \sum_i \alpha_i y_i x_i \tag{4}$$

The weight matrix is of size 1-by-2275. When plotted, the weight matrix resembles an average human eye, which proves the working of the trained classifier. The bias, b, was calculated by the kernel classifier during training. From the weight matrix and the bias, the confidence level of every point in the image was calculated using Eq. (5).

$$y_i(< w, x_i > + b) \geq 1 \tag{5}$$

The limited facial region where the search for both eyes is performed is in the face center where the eyes are located as per the human face golden ratio [12], illustrated in Fig. 5.

The search for eyes in step 3 is performed in a way similar to a moving fixed-sized window. This window SVM method for facial feature extraction has also been implemented by M. H. Nguyen, et al, at Carnegie Mellon University, U.S. [13]. In our case, the image is divided into sub-windows of size bigger than the 35-by-65 size of the training samples. In each window, the confidence level is calculated using the weight matrix of trained SVM. If the confidence level exceeds a set threshold, the eye is detected by the SVM. As a test result, an image with box displayed over the face is shown in Fig. 6. The confidence level, as displayed on the image, is 0.73888, which is above the threshold (set at 0.5). The box coordinates are [10.50 5.50 34.50 29.50]. Dividing the box width by half $(34.5 + 10.5)/2 = 22.5$ gives the exact position in between the two eyes, as shown in Fig. 7.

3. Classification phase

After the eyes are detected and tracked, labeling needs to be defined for eyeball movements. To make it easier for physically challenged people, three eyeball movements are defined. 'Center' position of the eyeball indicates that no key is to be

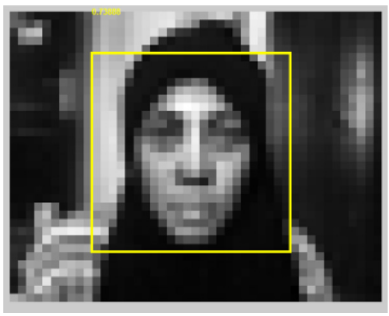


Fig. 6. Face detected in box using trained SVM.



Fig. 7. Middle of the box in SVM detected face matches exactly the face center.

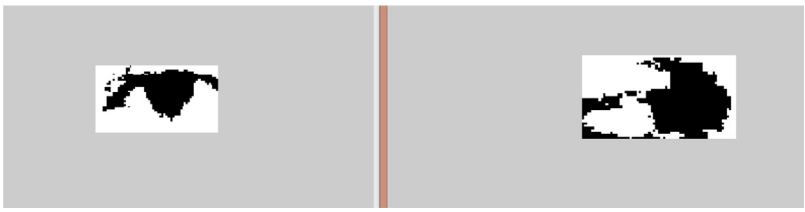


Fig. 8. Centre position of eye ball means no key pressed; side position means key pressed.

pressed, while ‘left’ and ‘right’ positions of the eyeball indicate that a key is pressed, as shown in Fig. 8. Thus, the scanning keyboard works by predicting the user’s eyeball positions. This is done in two steps. First, the method of dark pixel detection discussed under the tracking phase is implemented. Next, the estimation of the eyeball position is done by calculating the ratio of white to black pixels in different positions of the tracked eye to locate the eye pupil. This is exemplified in Fig. 9.

If the position of eyeball is detected as ‘left’ or ‘right’, the highlighted key on the keyboard is to be typed on the screen. This is because the designed keyboard works similar to the technique of Partner-Assisted Scanning. The latter technique is used for people with communication difficulties; where keys/words/sentences are highlighted consecutively. In other words, once the desired key is highlighted and the person wants to communicate that “word/sentence”, then a simple gesture (i.e., eyeball movement in our case) at that time would mean that this word/sentence is the desired one. We simulated this technique using computer screen through software, where a keyboard scans through each letter sequentially. This is illustrated in Fig. 10, where the yellow square moves through each letter, pauses for one second and then moves to the next letter and so on. When the desired letter is highlighted, the user gestures by moving her/his eyes to one side. When the system detects that the user’s eyeball has moved sideways, that highlighted letter is printed on the screen. The speed of the pause can be customized to match the pace of the user. This set up is thus projected as a technology assisted alternative to communicating with paralyzed people.

Phase 2 (tracking) and Phase 3 (classification) work independently. This is due to the single thread nature of the program used. In phase 2, eyes are tracked by the system, while in the phase 3 classification of eyeball movement is used to respond to sequential scanning of the keyboard letters. In other words, when eyes tracked in phase 2 indicate ‘left’ or ‘right’ eyeball

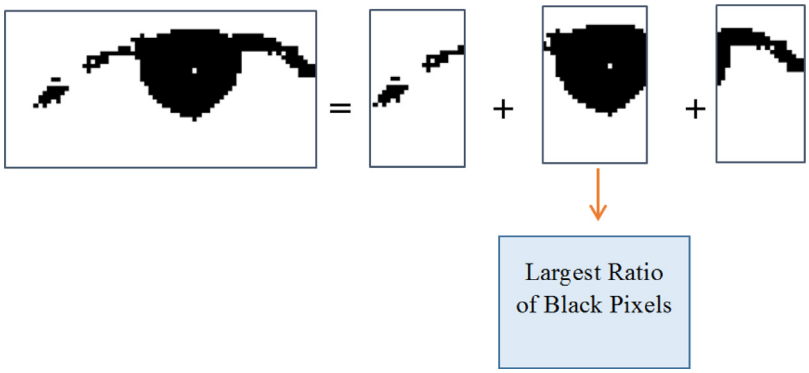


Fig. 9. Calculation of eye ball position done by calculation of black/white pixel ratio.

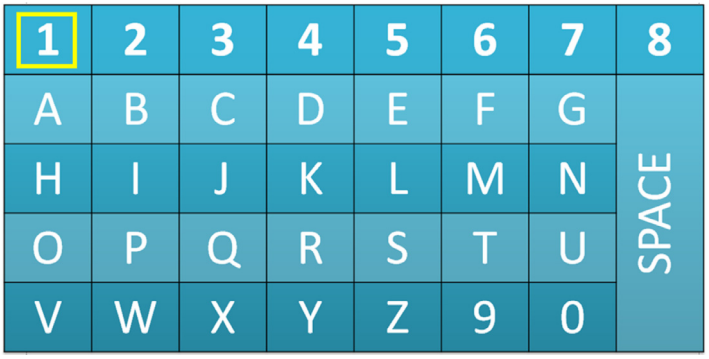


Fig. 10. Designed partner-assisted scanning keyboard.

Table 2
Testing results of tracking stage.

Test #	Results	Participant
Test 1		Female
Test 2		Male
Test 3		Male
Test 4		Female (child)
Test 5	Right eye and left eye tracked correctly	Female
Test 6		Female
Test 7		Female
Test 8		Female
Test 9		Female
Test 10		Male
Test 11		Male (child)

position, the key to being retrieved and typed on the screen is calculated by a third program with the help of a timer as in Eq. (6). With space counted 4 times (once at the end of every row, except for the first row), there's a total of 40 keys. With 40 letters existing on the keyboard, the entire keyboard is scanned in 40 s since the keyboard scans at a fixed time of one second per letter. Calculating the time elapsed indicates which particular key is currently scanned by the keyboard. Thus, in total, the PAS keyboard is designed to run three files simultaneously. This is enabled in programming such that the software used is a single thread and does not allow for multiple threads to run on a single program.

$$key = remainder\ of\ \frac{time\ elapsed\ in\ seconds}{40} \tag{6}$$

4. Simulations

In this section, testing of the entire system is reported: starting with eye tracking followed by a classification of eye gestures. During testing of the tracking stage, the following results were observed as shown in Table 2. The environment was indoor with incandescent bulbs since the system used by physically challenged people is typically indoor. Testing was

Table 3
Testing results of whole integrated system.

Test #	Results	Reason
Test 1 (female)	Key not selected	Eyes detected correctly as 'center'
Test 2 (male)	Key selected	Eyes detected correctly as 'side'
Test 3 (male)	Key selected	Eyes detected correctly as 'side'
Test 4 (female, child)	Key not selected	Eyes detected wrongly as 'center'
Test 5 (female)	Key not selected	Eyes detected correctly as 'center'
Test 6 (female)	Key not selected	Eyes detected correctly as 'center'
Test 7 (female)	Key selected	Eyes detected correctly as 'side'
Test 8 (female)	Key not selected	Eyes detected correctly as 'center'
Test 9 (female)	Key selected	Eyes detected correctly as 'side'
Test 10 (male)	Key selected	Eyes detected wrongly as 'side'
Test 11 (male, child)	Key not selected	Eyes detected correctly as 'center'
Test 12 (female, child)	Key not selected	Eyes detected correctly as 'center'
Test 13 (male)	Key selected	Eyes detected correctly as 'side'
Test 14 (male)	Key not selected	Eyes detected correctly as 'center'
Test 15 (male)	Key selected	Eyes detected correctly as 'side'
Test 16 (male)	Key not selected	Eyes detected correctly as 'center'
Test 17 (female)	Key not selected	Eyes detected correctly as 'center'
Test 18 (female)	Key not selected	Eyes detected correctly as 'center'
Test 19 (male)	Key selected	Eyes detected correctly as 'side'
Test 20 (male)	Key selected	Eyes detected correctly as 'side'
Test 21 (male)	Key selected	Eyes detected correctly as 'center'
Test 22 (female, child)	Key not selected	Eyes detected correctly as 'center'
Test 23 (male)	Key selected	Eyes detected correctly as 'side'
Test 24 (male)	Key selected	Eyes detected correctly as 'side'
Test 25 (male)	Key selected	Eyes detected correctly as 'side'
Test 26 (male)	Key not selected	Eyes detected correctly as 'center'
Test 27 (female)	Key selected	Eyes detected correctly as 'side'
Test 28 (male, child)	Key selected	Eyes detected correctly as 'side'
Test 29 (male)	Key not selected	Eyes detected wrongly as 'center'
Test 30 (male)	Key selected	Eyes detected correctly as 'side'

Table 4
Comparing proposed work with other works.

Comparison with other works		
	Accuracy	Number of samples
MAP training, SIFT, RANSAC, homography [6]	85% single eye detection, 45% double eye detection	30
Skin color segmentation, Circular Hough Transform [8]	70% single eye detection, 60% double eye detection	30
Proposed algorithm	90% double eye detection	30

conducted on eleven people from both genders with ages ranging from 6 to 51. The purpose of both genders and different ages is to detect any human face regardless of face variations, thus increasing the reliability of the system.

The integrated system (i.e., detection and tracking) was built by combining face detection, eye tracking, and classification. Testing of the whole integrated system was done indoors, under fluorescent light bulbs, with people of both genders aged from 6 to 51 from different cultural backgrounds with head orientated towards the camera with no tilt. The no head tilt and indoor set-up are ideal for paralyzed people who are the focus of this work. The test data can be regarded as general, but the system will produce similar results if tested again on people of any age group, gender, and appearance under similar circumstances. The results are shown in Table 3. Based on results in Table 3, the Success Rate (SR) of the eyeball tracking scanning keyboard was calculated as:

$$\frac{Number\ of\ correct\ display}{Total\ number\ of\ trials} = \frac{27}{30} \times 100\% = 90\% \tag{7}$$

This work can also be compared with other similar works like in [6,8]. The thirty images as shown in Table 3 were also used as input to algorithms reported in [6,8], and the comparative performance efficiency is illustrated in Table 4.

When comparing the proposed algorithm to [8], the former performs better since it achieves higher accuracy rates (i.e., 90% double detection to 60% double detection). While comparing the proposed algorithm to [6], the former achieves higher accuracy rates (i.e., 90% double detection to 450% double detection). Typical reasons behind comparative performance are not the only number of samples [14], though the number of test samples affects the reliability of the work. But, as in [14], the other factors also come into play when comparing works quantitatively. For instance, while both works (proposed algorithm and [6]) use an i7 core processor, yet in [6] images are of the high pixel resolution of 1944×1296, compared to the proposed algorithm that works on low resolution (i.e., 640×480) images. This parameter adds strength to our proposed algorithm.

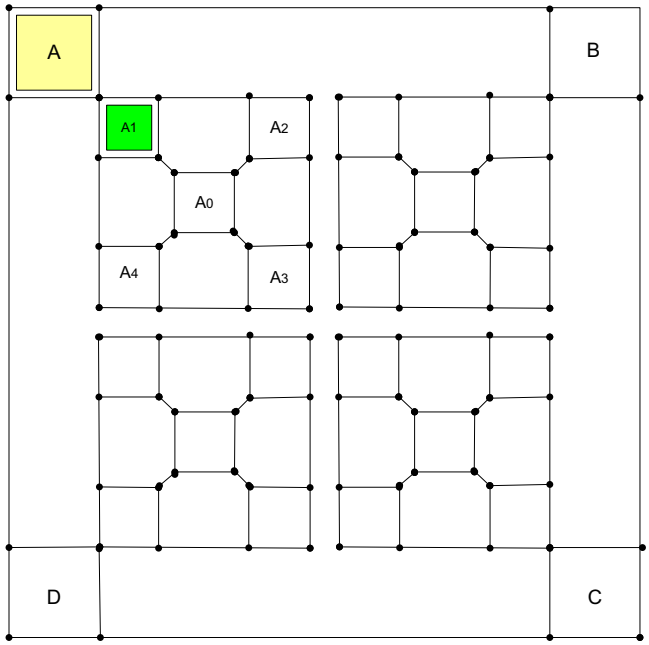


Fig. 11. Alternative PAS keyboard design.

For tracking stage, SVM-based approach proved effective with 100% accuracy compared to the morphological approach discussed in [15]. The reason behind this is that the edge detection used in [15] to detect the elliptical shape of the eyes failed in our work due to low-resolution images under room lightings. Such lighting is usually dim, and shadows tend to hinder the shape of the eyes in the captured images.

5. Benefits of the proposed system

Till today, solutions built are not up to the mark as they lack in enabling automated mobility, patient rehabilitation, entertainment, and technology-assisted communication. The proposed technology-assisted system can be extended, for example, in facilitating the mobility of the physically challenged person in moving the bed to an appropriate position or the person on the wheelchair to move it to a different direction, all in absence of the nurse. As an example, with appropriate design of the keyboard, the person (either on the bed or on the move) can communicate to the outside world by appropriately selecting an option on the keyboard to enable connecting to a restaurant website to order specific food, sports website to watch live streaming, etc. Furthermore, the system can also be used for gaming and exercise of nerves during the rehabilitation process, where patients with nerve problems would not only be able to play already existing games, but the application may work as 'mouse and click' needed for brain games. In this situation, the eyeball movement (up-down, left-right) works like mouse movement and blink of an eye as a click.

The testing keyboard was designed as simple as possible. The first advantage of the proposed system is that the scanning keyboard is extendable, as more buttons can be added. For example, the eye tracking scanning keyboard based on the technique of Partner-Assisted Scanning can be designed to detect a variety of different eye movements. The keyboard could display letters, sentences, objects, or web addresses. They can be designed in different layouts and languages. One of such layout as an alternative to the one shown in Fig. 10 is drawn in Fig. 11. The objective behind such a design is to support communication, mobility, entertainment, and patient rehabilitation.

In Fig. 11, the yellow square moves through each outer (corner) square with a clockwise cycle of four positions – by pausing for one second and then moving to the next letter position and so on. The speed of the pause can be customized to match the pace of the user and tracking the duration of the system. When the desired letter is highlighted, the user gestures by moving his/her eye ball to one side. When the system detects that the user's eyeball has moved sideways, it means that the option on the keyboard is selected and keyboard enters inner (smaller) five squares. Here, the green square moves in a clockwise motion (left-up corner, right-up corner, right-down corner, left-down corner) for one cycle and goes to the center square before it starts the inner cycle again. If selected, the center of each inner square-set is used to leave/skip inner squares and come back to respective outer yellow square. The displayed keyboard has four categories of functions mapping to A: Communication, B: Mobility, C: Rehabilitation, D: Entertainment, the example is shown in Table 5 and can be extended to support additional categories and/or functions.

The second advantage of the method proposed is that it uses a normal webcam that is affordable by all people and is typically found in every ordinary mobile phone. As such, this system could be installed on any mobile device. The use

Table 5
Keyboard squares and sample mapped functions.

Categories	Functions				
A (communication)	A_1	A_2	A_3	A_4	A_0
	Water	Lights on/off	Food	Washroom	Back to A
B (mobility)	B_1	B_2	B_3	B_4	B_5
	Position 1	Position 2	Position 3	Position 4	Back to B
C (rehabilitation)	C_1	C_2	C_3	C_4	C_5
	Play game 1	Play game 2	Play game 3	Play game 4	Back to C
D (entertainment, sports)	D_1	D_2	D_3	D_4	D_5
	Website 1	Website 2	Website 3	Website 4	Back to D

of alternative ways, such as a depth sensor camera or a more advanced camera will make the tracking code simpler and shorter. This is because the system will work with regard to the closest object as the user and thus no more complications in the code are required. However, use of an advanced camera implies that users need to buy additional hardware (since a depth camera is not as common as a webcam). This implies that there is a trade-off between a simple software solution with additional hardware and complicated software with no hardware requirements on existing mobile phones.

6. Discussions and conclusions

This paper presents the development of a simple low-cost scanning keyboard that works on detection of eyeball movements. The presented approach uses eye detection and tracking, followed by eyeball position classification. The developed keyboard was tested indoors, under fluorescent light bulbs with people of both genders aged from 6 to 51 from different cultural backgrounds with head oriented towards the camera. Based on testing conducted on thirty people, the accuracy achieved is 90% for double eye detection.

There are some remarks on the proposed eyeball tracking system. First, the presence of an object with the same color as the skin degrades the system performance and might cause inaccurate tracking of the eyes. Therefore, the background is preferred to be plain with a color different than that of the user's skin color. Secondly, in terms of face orientation, the system was designed to detect eyes that are aligned horizontally. The developed system allows the user to be seated anywhere between 60 to 180 cm away from the computer screen. In terms of head orientation, the skin color segmentation will detect faces tilted by any angle. However, the SVM was trained with eye samples looking straight into the camera. As such, a huge angle tilt of the user's head will mostly lead to false tracking by the camera, since the confidence level will not exceed the threshold set for the SVM to detect eyes. However, a small deviation of the head tilt of an angle of 0–15° along the horizontal line will not affect the system performance. Also, this degree can be controlled by the program code during SVM training.

In the future, this work can be improved further in many ways: by using high- resolution images to decrease false detection; by allowing further head orientations, as well as colored backgrounds. Also, depth camera implantation will definitely improve the results.

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Zeenat Al-Kassim completed her Bachelor's degree in Computer Engineering in 2013 from Prince Mohammad Bin Fahd University, Saudi Arabia. She joined UAE University in 2014 as Masters student in Electrical Engineering. She graduated in Spring 2016. Her research interests are in imaging and video applications.

Qurban A Memon graduated from University of Central Florida, Orlando, US with PhD degree in 1996. Currently, he is working as Associate Professor at UAE University. He has authored over ninety publications. His research interests are in the area of intelligent systems and security. He has served as a reviewer of many international journals and conferences.